

**torch.autocast wraps the forward pass and the loss ONLY - not backward, not the optimizer step**

### **Cast to BF16**

linear layers (matmuls)

attention  $QK^T$  and  $PV$

convolutions

**-> activations become BF16**

### **Left in FP32**

layer norm

softmax

the loss (cross-entropy)

**-> parameters stay FP32**

*schematic; matmul-like ops are cast, reduction-heavy ops are not*